
OenoBench: A Wine-Domain Benchmark for Knowledge-Grounded Evaluation of Large Language Models

Nikita Khudov, DipWSET
StrategAI
nikitahudov@gmail.com

Abstract

We introduce **OenoBench**, a wine-domain knowledge benchmark of **3,266 multiple-choice questions** across six pillars (regions, grape varieties, viticulture, winemaking, producers, business) and four difficulty tiers. The corpus is built from **38,104 atomic, source-anchored facts** extracted by 35 provenance-verified scrapers from government registries (INAO, TTB, OIV), peer-reviewed journals, and Wikipedia/Wikidata. Our methodological contribution is an *LLM-driven pipeline* in which language models reformat verified facts and audit the result, but never serve as the source of truth: every claim traces to a URL, every question is generated by one of five strategies across five generator families, and every question is scored by a nine-agent audit calibrated against a human gold sheet via Cohen’s κ . Evaluating sixteen frontier configurations, we find: (i) overall accuracy spans 53%–84%, led by **o3 at 83.6%**; (ii) reasoning-mode lift concentrates in **DeepSeek R1 (+6.8 pp)** and is absent in Claude Opus and Gemini Pro; (iii) Anthropic shows **+9 pp self-preference** on its own questions while Google shows **−8 pp** inverse preference; (iv) frontier open-weight models share the cost-vs-accuracy Pareto frontier with proprietary reasoning models; and (v) every config gains **≈33 pp** on closed-book-solvable items, revealing a parametric-recall ceiling that only the contextual slice avoids. We release corpus, audit findings, human-review app, and construction code under CC-BY-SA-4.0.

1 Introduction

Large language models are increasingly evaluated on knowledge-intensive benchmarks, but the most widely-used resources — MMLU [6], GPQA [16], and their successors [19, 21, 15] — share three properties that limit their usefulness as the field matures: (i) they aggregate breadth at the expense of *depth* within any single discipline, (ii) they are typically authored or curated by a small team using a single generation pipeline, exposing them to systematic blind-spots and stylistic regularities that frontier models can learn to exploit [14, 20], and (iii) the questions are increasingly suspected of contamination through web-scale pre-training corpora [17, 13, 3]. As models grow more capable, the marginal value of a benchmark depends increasingly on *how* it was constructed, not only on what it contains.

We argue that domain-specialised benchmarks built around bodies of knowledge that are *externally validated by humans* — through certification, regulation, or peer-reviewed practice — offer a useful complement to broad knowledge benchmarks. Examples already exist in medicine [9, 10], law [5], and finance [8]. We extend this line of work to a domain that has so far received little attention from the benchmarking community, despite being unusually well-defined: **wine**.

Why wine. Wine is a multi-disciplinary subject that integrates plant biology, geology and climate science, organic chemistry, food microbiology, sensory neuroscience, regulatory law, and business economics. It is one of very few domains for which (a) authoritative governmental and inter-governmental sources publish structured taxonomic data (e.g. INAO, OIV, TTB, regional consortia), (b) university research groups (notably UC Davis, Geisenheim, Bordeaux Sciences Agro) produce open scholarly literature, and (c) professional certification bodies define a graded competence ladder (WSET Levels 1–4, Court of Master Sommeliers, Institute of Masters of Wine) whose published syllabi can be used to ground what “expert” should mean. Together these properties make it possible to build a benchmark in which every question is traceable to a verifiable source, while still spanning the recall–reasoning–application spectrum that distinguishes a strong taster, winemaker, or buyer from a memorised glossary.

Contributions.

1. **A 3,266-question wine benchmark** across six pillars and four difficulty tiers, built from 38,104 atomic provenance-tagged facts (Section 3).
2. **An LLM-driven, fact-grounded generation pipeline** with five strategies $\times$ five generator models, per-strategy quotas, and an explicit closed-book pre-screen — LLMs reformat verified facts, never supply them.
3. **A nine-agent automated audit** (Section 4) calibrated against a human gold sheet via Cohen’s κ ; the audit removed 341 questions and re-labelled difficulty on 1,259, and surfaced one agent (closed-book solvability) that over-reports leakage relative to humans, which we keep with explicit disclosure rather than drop.
4. **A bias-aware evaluation across sixteen frontier configurations** (Section 5) reporting overall accuracy, reasoning-mode lift, a Self-Preference Score, cost-efficiency Pareto, and a closed-book vs. source-grounded contrast that isolates parametric recall from contextual reasoning.

OenoBench, the construction and audit pipelines, the multi-agent audit findings, and the human-review web application are released under CC-BY-SA-4.0; see Appendix E for dataset and code URLs and the double-blind anonymisation map.

2 Related Work

Broad knowledge benchmarks. MMLU [6], BIG-Bench [19], AGIEval [21], GPQA [16], HELM [12], and Humanity’s Last Exam [15] measure factual breadth and reasoning across many subjects. As headline scores have saturated, attention has shifted toward harder reasoning benchmarks and toward auditing the construction process. Our work is complementary: rather than maximising difficulty across many disciplines, we maximise depth within a single externally-validated knowledge domain, and treat *construction-time bias control* as a first-class methodological contribution rather than a downstream decontamination step.

Domain-specific benchmarks. MedQA [9], PubMedQA [10], LegalBench [5], and FinanceBench [8] demonstrate the value of grounding evaluation in a regulated profession with clear sources of truth. We adopt their organising principle — expert syllabus as taxonomy, source-traceable facts as backbone — and add multi-model question generation, an automated multi-agent audit, and self-preference reporting as additional bias controls. To our knowledge, OenoBench is the first such benchmark for the wine domain.

LLM-as-judge bias and self-preference. A growing literature documents that models tend to favour outputs stylistically similar to their own [14, 20, 18]. This biases benchmarks where the same model generates and grades. We mitigate this in two ways: by distributing generation across five models (Claude, GPT, Gemini, Llama, Qwen) plus deterministic templates, and by reporting per-model self-vs-other accuracy gaps as a first-class diagnostic (Section 5.3).

Benchmark contamination and solvability. Pre-training exposure inflates apparent capability [17, 13, 3]. We mitigate contamination in two ways: (i) generating questions from atomic facts rather than copying source text (a fact-echo agent flags any question whose longest common substring with

the source exceeds 65%), and (ii) running an explicit *closed-book solvability* pre-screen and audit, allowing us to report a separate per-config accuracy on world-knowledge-solvable items (Section 5.5). We further show that the same audit, when calibrated against a human gold sheet, reveals that LLM judges themselves over-report leakage by an order of magnitude — a finding with broader implications for the LLM-as-judge paradigm.

Wine in NLP. Wine has previously appeared in NLP only as a substrate for review-text generation, recommendation [2], taxonomy learning [11], and judge-reliability studies [7]. OenoBench is the first benchmark that targets *structured factual knowledge* of viticulture, winemaking, and wine regions evaluated against frontier LLMs.

3 Dataset Construction

OenoBench is constructed by an LLM-driven pipeline (Figure 1) in two stages: (1) a *fact-collection* stage that builds a corpus of atomic, provenance-tagged statements from authoritative sources, and (2) a *question-generation* stage that converts those facts into benchmark questions using five complementary strategies and five generator models. The pipeline is designed around a single methodological commitment: large language models are used as *rephrasers and auditors*, never as the source of truth. Every fact in the corpus traces to a URL with a tier-of-authority label, every question is anchored in one or more such facts, and every question is scored by a nine-agent automated audit (Section 4) before release.

3.1 Domain taxonomy

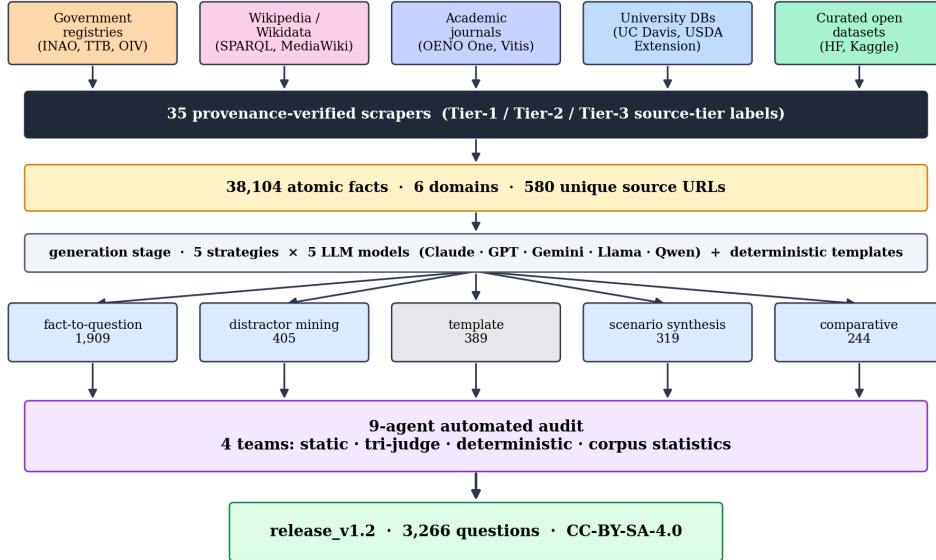


Figure 1: End-to-end OenoBench pipeline: authoritative sources → 35 scrapers → atomic facts → closed-book gate → nine-agent audit → release.

We organise wine knowledge into six *domain pillars*, chosen to align with the WSET Level 3 and Diploma syllabi: (i) **wine regions** (appellations, classifications, geography, climate, soils); (ii) **grape varieties** (ampelography, parentage, regional distribution, sensory profile); (iii) **viticulture** (vineyard practice, training systems, pests and diseases, vintage effects); (iv) **winemaking** (vinification choices, fermentation, ageing, faults); (v) **producers** (estates, négociants, co-operatives, key houses); and (vi) **wine business** (markets, regulation, distribution, economics). Each domain has a separate target share informed by syllabus weighting and source availability; the realised distribution is reported in Table 1.

3.2 Source tiering, licensing, and non-fabrication

Every fact is tagged with one of three source tiers, mirroring evidence-based-medicine practice:

- **Tier 1 (official, 19.6%):** government registries and inter-governmental bodies (INAO, OIV, TTB, regional consortia, the UC Davis AVA database).
- **Tier 2 (authoritative, 76.6%):** Wikipedia (34.3% of facts), Wikidata SPARQL (31.0%), wine-body and consortium websites, peer-reviewed journals (*OENO One*, *Vitis*, *Australian Journal of Grape and Wine Research*).
- **Tier 3 (reliable, 3.8%):** curated open datasets on HuggingFace and Kaggle, secondary reference databases.

Licensing and non-fabrication. Every source is public-record or released under a licence that permits scraping and redistribution: Tier-1 government data under explicit open-data licences (INAO Licence Ouverte, OIV, UC Davis AVA, TTB public-domain), Tier-2 Wikipedia/Wikidata under CC-BY-SA / CC0 and journals (*OENO One*, *Vitis*, *AJEV*) open-access; the full per-source licence table is in Appendix E. A central design commitment is that *no fact in the corpus comes from an LLM’s internal knowledge*: every fact is extracted by a source-specific scraper, stamped with source URL, retrieval timestamp, and tier label before insertion, and the fact-insertion utility rejects records without a verified source-URL trace. We emphasise this because LLM-generated content masquerading as scraped data is a contamination vector that conventional dataset documentation does not detect.

Table 1: Atomic-fact corpus: 38,104 facts from 35 scrapers, six domain pillars, three source tiers.

Domain pillar	Facts	Share	Tier 1	Tier 2/3
Wine regions	18,943	49.7%	23.4%	76.6%
Producers	6,215	16.3%	9.1%	90.9%
Grape varieties	5,959	15.6%	18.8%	81.2%
Viticulture	3,635	9.5%	12.0%	88.0%
Wine business	1,985	5.2%	31.4%	68.6%
Winemaking	1,367	3.6%	16.7%	83.3%
Total	38,104	100.0%	19.6%	80.4%

3.3 Atomic fact extraction

Source pages are decomposed into atomic facts by a five-stage pipeline (Figure 6, Appendix B.1): (1) sentence splitting at conjunctions and clause boundaries, (2) reference resolution (pronouns and demonstratives are replaced with their entity referents from article context), (3) domain classification into one of the six pillars, (4) a length-and-predicate validator that rejects facts shorter than five words, longer than fifty words, missing a verb, or carrying dangling references, and (5) an on-topic filter parameterised by region-specific keyword sets to prevent cross-contamination (e.g. Austrian content appearing in a Bordeaux scraper). Wikidata SPARQL queries use direct country relations (P17) rather than transitive administrative parents (P131*); the latter caused severe cross-region leakage in early scrapers and is documented as a cautionary case in Appendix B.

Facts are required to be (a) *atomic*, one assertion per sentence; (b) *entity-tagged*, with subject and object linked to canonical IDs in the wine knowledge graph; and (c) *source-faithful*, a paraphrase of the original, never a verbatim copy. This last constraint is enforced both at extraction time and at audit time (Section 4, agent A3).

3.4 Question generation: five strategies, five models

A single-strategy, single-model benchmark inherits the blind-spots of its pipeline. We mitigate this by combining *five strategies* and *five generator models*, with per-strategy and per-model quotas enforced by an orchestrator that resumes safely across runs. Strategies are:

- **Fact-to-question (58.4%, 1,909 questions):** the LLM rewrites a single verified fact as a multiple-choice question; preserves grounding but tends toward recall.

- **Distractor mining (12.4%, 405):** confusable-entity sampling produces wrong answers that share the right answer’s category and dimension, raising plausibility and forcing fine-grained discrimination.
- **Template (11.9%, 389):** forty-five deterministic parameterised templates across the six pillars; pure entity substitution, zero LLM creativity — a baseline against which neural generation is measured.
- **Scenario synthesis (9.8%, 319):** coherent fact clusters are converted into applied multi-fact reasoning prompts (e.g. a sommelier service decision, a winemaker blending decision, a viticulturist vintage call). Domain-specific scenario types prevent persona-content mismatch.
- **Comparative (7.5%, 244):** entity-affinity scoring pairs related entities (e.g. two Burgundy premiers crus, two Rhône grapes) and asks “which differs in X” or “what do both share”.

The five generator models are Claude Opus 4.7, GPT-5, Gemini 2.5 Pro, Llama 3.1 405B, and Qwen 3.5 235B, each accessed through a unified OpenRouter client with consistent temperature and top- p . The realised generator distribution across the released corpus is **Qwen 20.4%, Llama 19.3%, Claude 19.0%, ChatGPT 16.6%, Gemini 12.9%, deterministic templates 11.9%** (computed directly from the 3,266 released questions). The orchestrator allocates generation share dynamically based on per-pilot audit pass rates — generators whose questions pass audit reliably get a larger share in subsequent rounds — subject to a hard **per-generator cap of 21%** to keep any single model from dominating the corpus.

3.5 Final corpus: release_v1.2

After generation, audit, and post-eval review, the released corpus `release_v1.2` contains **3,266 questions**. The distribution along the domain and difficulty axes is:

Table 2: Composition of the released corpus.

Domain	N	%	Difficulty	N	%
Wine regions	1,108	33.9%	L1 (entry)	694	21.2%
Grape varieties	766	23.5%	L2 (intermediate)	894	27.4%
Producers	515	15.8%	L3 (advanced)	678	20.8%
Viticulture	502	15.4%	L4 (expert)	1,001	30.6%
Wine business	250	7.7%			
Winemaking	187	5.7%			
Total	3,266	100%	Total	3,266	100%

`release_v1.2` ships under CC-BY-SA-4.0 on HuggingFace with a Croissant manifest [1] and a full Datasheet [4] in Appendix A.

4 Multi-Agent Quality Audit

The audit pipeline runs every generated question through nine automated agents organised into four teams. Each agent emits a per-question signal in $\{\text{PASS}, \text{WARN}, \text{FAIL}\}$ and signals are calibrated against a human-reviewed gold sheet using Cohen’s κ . Agents whose human agreement falls below threshold are downweighted to advisory-only, on the principle that an automated judge whose verdict humans cannot reproduce should not gate question admission. A separate human-review web application (Appendix D) collects the gold-sheet ratings from WSET-Diploma reviewers and is itself part of the released artifact set. A team-by-team architecture diagram (Figure 5, Appendix C) shows how candidate questions flow through the four teams to the drop / relabel / keep verdict.

4.1 Team architecture

Of the 14 agents, **nine are always-run** on every candidate question; the remaining **five are escalation-gated** and are invoked only when an upstream agent flags a corpus-level concern. This deferred-execution policy keeps routine audit cost bounded ($\sim$ \$76 for the full corpus audit reported here, 5h 22m wall) while preserving full coverage when a defect class is suspected. The full audit code, prompts, and thresholds are released alongside the corpus (Appendix B).

Table 3: The 14 audit agents across four teams. ALWAYS: run on every question. ESC: escalation-gated.

Team	Agent	Run	Function
A. Static	A1 LexicalHygiene	ALWAYS	Regex sweep for vague phrasing, marketing language, meta-questions.
	A2 BiasStats	ALWAYS	χ^2 on correct-answer position; Mann-Whitney U on correct-vs-distractor length.
	A3 FactEcho	ALWAYS	Longest-common-substring ratio between question text and source fact (FAIL at $LCS \geq 0.65$).
	A4 TemplateFingerprint	ALWAYS	POS-bigram logistic regression detector for template-induced stylistic regularities (held-out AUC 0.84).
B. Tri-judge	B1 TriJudgeAnswer	ALWAYS	Claude / GPT / Gemini panel reads question + source fact and votes on the marked-correct option.
	B2 ClosedBookSolvability	ALWAYS	Same panel, source fact <i>withheld</i> : can the question be answered from world knowledge alone?
	B3 UbiquityRisk	ALWAYS	Internationally-grown grape stem $\times$ region-class answer $\Rightarrow$ ambiguity flag.
	B4 Ambiguity	ESC	Tri-judge re-read flags questions with >1 defensible answer; invoked when B1 dissent rate elevated.
	B5 VerifierSkip	ESC	Self-consistency probe across re-prompted panels; invoked on disputed B1 verdicts.
C. Determin.	C1 DistractorDifficulty	ESC	Embedding-distance distractor analysis; invoked when A2 reports length skew.
	C2 CategoryLeak	ALWAYS	Wine-type distractor validation; flags red/white/sparkling category mismatches.
	C3 SourceSwap	ESC	Substitutes alternate source facts to test answer stability; invoked on B1/B2 disagreement.
	C4 DifficultyAudit	ALWAYS	Gemini Pro re-rates difficulty; FAIL/WARN when delta ≥ 2 from assigned.
D. Corpus stats	D1 SelfPreference	ALWAYS	Per-generator-family own-vs-other accuracy on a held-out probe; corpus-level signal.
	D2 DedupCalibration	ESC	Near-duplicate audit at cosine ≥ 0.92 ; invoked on suspicious cluster signals.
	D3 SkewAudit	ALWAYS	Country / sub-domain over-representation χ^2 .

4.2 Gold-sheet calibration

A human gold sheet of 50 stratified questions per release is rated by a WSET-Diploma reviewer through the released human-review web app (Appendix D). **15+ gold-sheet rounds** have been completed across preliminary releases (audit pilots v1–v16) so that the agent thresholds and κ values reported here represent calibration aggregated over an iterative cycle, not a single batch. Each round rates against eight rubrics: `answer_correct`, `distractors_plausible`, `not_ambiguous`, `source_faithful`, `needs_source`, `no_vague_language`, `labels_correct`, and `verbatim_copy`. For each automated agent we compute Cohen’s κ against the corresponding human rubric on the gold sheet. **Agent signals with $\kappa < 0.6$ are downweighted to advisory-only and excluded from gating decisions.**

The most consequential calibration result concerns agent **B2 ClosedBookSolvability**: against the human reviewer, who solved only $\approx 12\%$ of the gold-sheet questions without seeing the source fact, the LLM tri-judge panel reported $\approx 83\%$ solvability, yielding $\kappa \approx 0.007$. We interpret this not as a defect of the benchmark but as a property of the *evaluator*: frontier LLM judges have absorbed enough wine knowledge during pre-training to over-attribute “world knowledge” to questions that humans find non-trivial. We therefore retain the 1,601 B2-flagged questions in the released corpus, document the finding in the datasheet, and report a closed-book vs. source-grounded contrast in Section 5.5 that turns the audit’s blind spot into a calibrated diagnostic of memorisation reliance.

4.3 Release-cycle results

The 9-agent audit on `release_v1.1` (3,670 candidate questions) produced the verdicts summarised in Table 5 (Appendix B.5): A1, A3, B1, C2 and B3 between them flagged 341 distinct questions for DROP, while C4 re-labelled difficulty on 1,259 questions and B2 flagged 1,601 as closed-book-solvable but kept them with disclosure.

After drops, a follow-up post-evaluation pass (Section 5 and Appendix C) audited the 97 questions that all 16 evaluation configs answered incorrectly: 54 were removed as defects (wrong ground truth,

all-correct options, duplicate options), 9 were dropped on borderline-review and the remaining 34 retained as legitimately hard. The final released corpus is 3,266 questions.

The C4-driven difficulty re-label was decisive: it shifted the corpus from 14% L3+L4 items (the original generator-assigned distribution) to **51% L3+L4** (Table 6, Appendix B.5), bringing the hardest-tier counts above the entry-tier counts and substantially sharpening difficulty-stratified analyses (Section 5.1).

5 Evaluation

We evaluate **16 frontier configurations** on the 3,266-question release in a single end-to-end run. The slate combines six within-family cost pairs (Claude Opus 4.7 vs. Claude Haiku 4.5, Gemini 2.5 Pro vs. Gemini 2.5 Flash, GPT-5 vs. GPT-5-mini, Llama 3.3 70B vs. Llama 3.1 8B, Qwen 2.5 72B vs. Qwen 2.5 7B), four reasoning-mode configurations (o3, DeepSeek R1, Gemini 2.5 Pro thinking, Claude Opus 4.7 thinking), and two additional standard models (DeepSeek V3, Mistral Large). Every config sees the same 3,266 questions, with single-letter output (A–D) at `max_tokens=5` and a five-stop fallback. We organise the evaluation around five questions, one per subsection.

5.1 Overall capability ranking

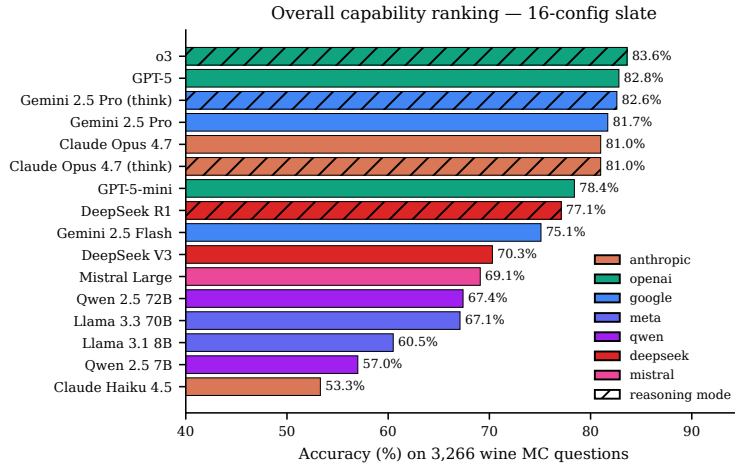


Figure 2: Overall accuracy on the 3,266-question corpus across 16 configurations. Reasoning-mode configs are hatched.

OenoBench cleanly stratifies the slate into three bands (Figure 2): **frontier** ($\geq 80\%$, led by **o3** at **83.6%**, with GPT-5, Gemini 2.5 Pro, and Claude Opus 4.7 clustered within 3 pp); **mid-tier** (mid-60s to high-70s); and **small-tier** (low-50s to low-60s). No config performs at random; the best-vs-worst spread is 30 pp, comparable to GPQA [16] but on a domain corpus $30\times$ smaller and fully provenance-tagged. Difficulty-stratified accuracies (Appendix C, Table 9) confirm a clean L1→L4 gradient: 93.6% on L1 down to **58.7% on L4 expert items**, with frontier configs holding 68–71% on L4 and small configs at 39–45%.

5.2 Reasoning-mode lift

Reasoning-mode lift is concentrated in a single family (Figure 9, Appendix C). Only **DeepSeek R1 vs. V3** (+6.8 pp [+4.6, +8.8]) has a CI excluding zero; the three frontier pairs (o3 vs. GPT-5, Gemini Pro thinking vs. Pro, Claude Opus thinking vs. Opus) are all statistically indistinguishable from zero. We attribute the asymmetry to headroom: V3 leaves substantial parametric ceiling at 70.3% which chain-of-thought can recover, while frontier models are already near their recallable ceiling. Per-difficulty breakdowns (Appendix C, Table 10) show the frontier reasoning configs gaining at most 1–2 pp at L4 and flat elsewhere, making the cost-vs-lift trade unfavourable except for DeepSeek (Section 5.4).

5.3 Self-preference bias

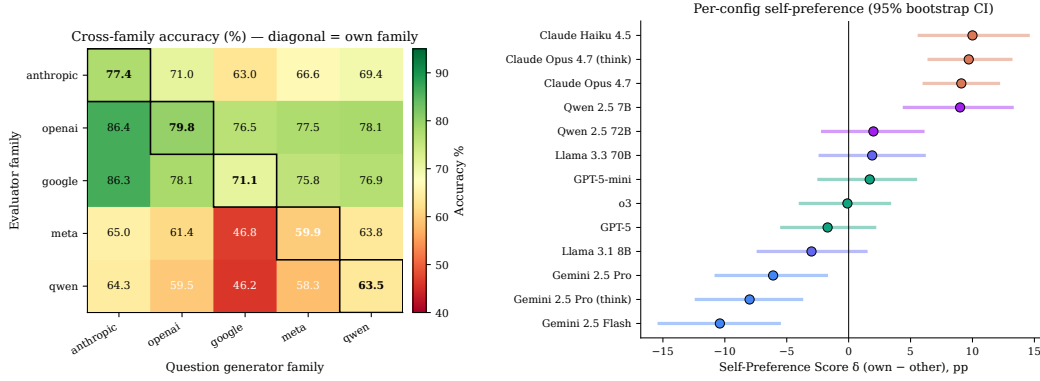


Figure 3: Self-preference bias. *Left*: cross-family accuracy matrix (rows = evaluator, columns = generator). *Right*: per-config SPS $\delta = \text{Acc}(\text{own}) - \text{Acc}(\text{other})$ with 95% bootstrap CI.

We compute the *Self-Preference Score (SPS)* per config m as $\text{SPS}(m) = \text{Acc}(m \mid Q_{\text{own}}) - \text{Acc}(m \mid Q_{\text{other}})$ where Q_{own} collects questions generated by m ’s family and Q_{other} its complement. A pipeline that successfully neutralises generator fingerprints should produce $|\text{SPS}|$ near zero.

The result (Figure 3) is family-dependent: **Anthropic** clusters at +9 to +10 pp (all CIs above zero), **OpenAI** is statistically zero, and **Google** shows an unexpected *negative* −6 to −10 pp cluster. The cross-family matrix (Figure 3, left) traces the asymmetry: Google-generated questions are uniformly harder for everyone, while Anthropic-generated questions are uniformly easier for Anthropic only. We interpret the Anthropic gap as a residual stylistic fingerprint that pre-release paraphrase passes did not neutralise, and the Google gap as harder-by-phrasing rather than by leak; both are first-class disclosures with the released corpus.

5.4 Cost efficiency

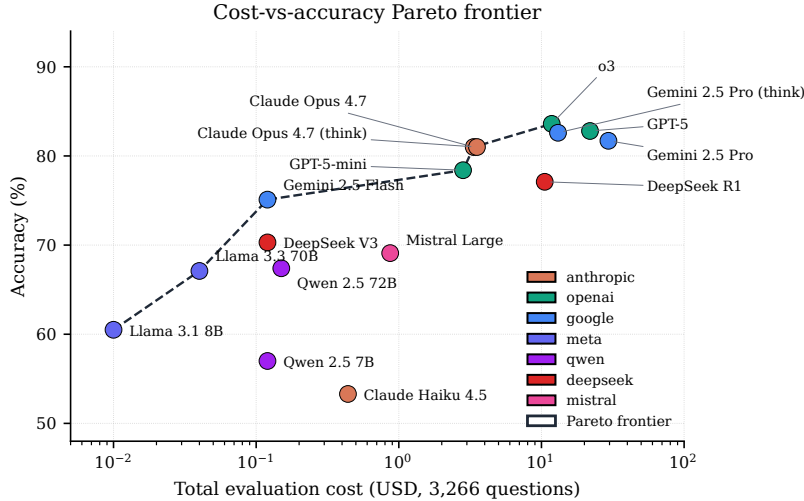


Figure 4: Total OpenRouter cost (log scale) versus accuracy on the full 3,266-question slate. Family colours match Figure 2; the Pareto frontier (red dashed) illustrates the cost-vs-quality trade-off.

The slate spans nearly four orders of magnitude in evaluation cost (Figure 4). Five configs sit on the Pareto frontier spanning the cost-quality spectrum: **Llama 3.1 8B** (60.5 % / \$0.01), **Gemini 2.5 Flash** (75.1 % / \$0.12), **GPT-5-mini** (78.4 % / \$2.82), **Claude Opus 4.7** (81.0 % / \$3.35), and **o3** (83.6 % / \$11.80). The two most expensive configs (Gemini 2.5 Pro \$29.47, GPT-5 \$21.90)

are dominated by cheaper Pareto neighbours: their extra spend buys output tokens, not better wine answers. Reasoning-mode lift (Section 5.2) yields a Pareto win only for DeepSeek R1, which is itself dominated by GPT-5-mini. The efficient boundary therefore favours small proprietary plus a single reasoning model over reasoning-everywhere or scale-everywhere strategies.

5.5 Closed-book vs. source-grounded performance

The audit’s B2 agent reported $\sim 83\%$ closed-book solvability against the human reviewer’s $\sim 12\%$ (Section 4.2) — itself a measurement of what frontier LLMs already *know* about wine. We exploit it by partitioning the corpus into the 1,601 B2-flagged questions and the 1,665 contextual / source-grounded questions, and reporting per-config accuracy on each slice (Figure 11, Appendix C.5).

The contrast is the largest single effect we observe. The mean per-config closed-book minus contextual gap is **+32.6 pp** (range +26.6 to +39.6 pp, every CI above zero). Two readings are mutually consistent: the B2-flagged slice overlaps heavily with parametric wine knowledge acquired in pre-training, while the contextual slice is where wine knowledge has to be reasoned through the source fact — **Claude Opus 4.7 and o3 hold $\sim 70\%$ on the contextual slice, small models drop to $\approx 45\%$** , and the inter-config spread widens from 27 to 33 pp. Downstream users can therefore evaluate on the contextual slice for reasoning-with-context, or on the union for total wine capability; the contextual slice is the more discriminative test, while the B2-flagged slice is a calibrated indicator of how much apparent “wine knowledge” is parametric recall.

6 Applications and Future Work

OenoBench is a benchmark first, but the construction pipeline behind it is intended to be reusable. We outline three threads that build on the released artifacts and methodology.

6.1 Closing the wine-knowledge gap in low-cost models

The cost-vs-accuracy frontier (Section 5.4) shows small open-weight models trailing proprietary frontier models by 20–30 pp here, where the gap on broad benchmarks like MMLU is only 10–15 pp. The 38,104-fact corpus is a natural *supervised fine-tuning* target via a two-stage recipe: (i) instruction-style fine-tuning on a held-out quartile of question/source-fact pairs, teaching source-grounded reasoning; and (ii) LoRA adaptation on the full 38,104-fact corpus, exposed by the release schema as fact-question-source triples. Both run on consumer GPUs without proprietary weights or pre-training compute. The contextual slice (Section 5.5) is the harder-to-game evaluation surface, so measured gain is a faithful proxy for genuine wine reasoning. We expect a Llama-3.3-70B-class adaptation to recover much of the GPT-5-mini gap at $\approx 1\%$ of proprietary inference cost, opening cost-sensitive deployment paths for sommelier-tooling and wine-education applications.

6.2 Case-based evaluation with industry partners

Multiple-choice tests have ceilings: a perfect OenoBench score does not establish that a model can act usefully in real workflows. We are scoping an **applied case-based dataset** with industry partners along three personas:

- *Sommelier* $\rightarrow$ *customer service*: consumer pairing prompts rated by certified sommeliers on recommendation, justification, and substitutions.
- *Winemaker* $\rightarrow$ *blending decisions*: vintage write-ups, lab analyses, and regulatory constraints in context; outputs rated on technical correctness, regulatory compliance, and stylistic intent.
- *Viticulturist* $\rightarrow$ *vineyard decisions*: weather, soil/disease, canopy/yield records; canopy, irrigation, harvest-timing, and pest-control recommendations rated against decisions actually taken in the same vintage.

This requires expert annotation of free-form outputs, but the OenoBench pipeline (multi-strategy generation, multi-agent audit, gold-sheet calibration) and the IRR-aware human-review app (Appendix D) carry over. We treat OenoBench as the *factual-recall floor*: factual knowledge must be near-saturated before applied reasoning can be validly measured.

6.3 Generalising the LLM-driven scrape-to-audit pipeline

The pipeline elements — (i) tier-of-authority source taxonomy, (ii) atomic-fact extraction with paraphrase enforcement, (iii) multi-strategy multi-model generation with per-model quotas, (iv) automated multi-agent audit calibrated against humans by κ , and (v) closed-book pre-screen for parametric leakage — are domain-agnostic, implemented as separable modules. Swapping the source-tier configuration and domain taxonomy is sufficient to retarget. Three near-term targets share wine’s structural features: *horticulture/agronomy* (regulated appellations, peer-reviewed journals, certification ladders); *regulated medicine* (clinical guidelines and drug formularies — the strongest tier-of-authority structure of any field, where current benchmarks rely on board-exam questions misaligned with practice); and *financial regulation* (FASB, IFRS, jurisdictional filings; CFA/CPA ladders). In each case the released code reduces a months-long expert-authoring effort to a weeks-long curation effort, given careful tier-of-authority configuration.

7 Limitations

Snapshot vs. moving target. Wine regulation, classifications, and producer ownership change on a multi-year cadence (INAO appellation revisions, DOCG promotions, estate sales). The released corpus is a snapshot dated 2026-04-01 and will require periodic re-extraction; we release the scrapers and provenance metadata so this is mechanically reproducible.

Closed-book audit calibration. The B2 ClosedBookSolvability agent shows $\kappa \approx 0.007$ with humans on the gold sheet (Section 4.2). Rather than drop the affected questions, we keep them with explicit disclosure and report the closed-book vs. contextual contrast in Section 5.5. Downstream users who prefer a strictly source-grounded subset can use the contextual slice; those who want pure recall can use the closed-book slice. The documentation is a load-bearing part of the dataset, not a footnote.

Self-preference is not fully neutralised. The Anthropic +9 pp positive SPS (Section 5.3) implies the multi-model generation strategy did not fully neutralise stylistic fingerprints. We disclose per-config SPS as a first-class diagnostic rather than removing the affected items, but downstream comparisons between Anthropic and other-family models on the full corpus should be treated with the SPS in mind.

Acknowledgments and Disclosure of Funding

This work was independently funded by StrategAI. We thank the open contributors to Wikipedia, Wikidata, INAO, OIV, TTB, UC Davis, USDA Extension, and the wine consortia and academic journals (*OENO One*, *Vitis*, *AJEV*) whose work makes a provenance-grounded benchmark of this scale possible. We thank the OpenRouter team for the unified-API gateway used during construction and evaluation.

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A Datasheet for OenoBench

We follow the datasheet template of Gebru et al. [4].

A.1 Motivation

For what purpose was the dataset created? To evaluate factual knowledge of large language models on a multi-disciplinary, externally-validated domain (wine), and to provide a benchmark whose construction includes explicit bias controls (multi-model generation, multi-agent audit, self-preference scoring, closed-book vs. source-grounded partition).

Who created the dataset? The dataset was created by Nikita Khudov (StrategAI) for the NeurIPS 2026 Evaluations & Datasets Track. The lead author holds the WSET Diploma in Wines (the highest pre-Master-of-Wine qualification of the Wine & Spirit Education Trust) and produced the gold-sheet ratings used to calibrate the audit agents.

Funding. Independently funded by StrategAI. No external research grants were received. Total OpenRouter API cost across the entire project — generation pilots, audit cycles, and the final 16-config evaluation — was **\$783**, paid out-of-pocket.

A.2 Composition

What do the instances represent? Each instance is a multiple-choice benchmark question with: question text, 4 options with marked-correct answer, supporting fact(s) with source URL and tier label, generator model identifier, generation strategy, audit-agent verdicts (per-agent pass/warn/fail signals), and (for the gold-sheet subset) human-reviewer ratings on eight rubrics.

How many instances are there? **3,266** questions in `release_v1.2`. By strategy: fact-to-question 1,909, distractor mining 405, template 389, scenario 319, comparative 244. By difficulty (post-relabel): L1 694, L2 894, L3 678, L4 1,001. By domain: wine regions 1,108, grape varieties 766, producers 515, viticulture 502, wine business 250, winemaking 187. The corpus is built from 38,104 atomic facts across 35 scrapers and 580 unique source URLs.

What data does each instance consist of? Structured Postgres records mirroring the schema documented in Section 3 and Appendix B.

Recommended data splits. The dataset ships as a single test split (this is an evaluation-only benchmark). Two analytic partitions are documented and recommended for use: (i) the *closed-book* slice (1,601 questions tagged `closed_book_solvable`) and the *contextual* slice (1,665 remaining questions), partitioned by audit agent B2; and (ii) per-generator held-out splits for self-preference analysis (Section 5.3), defined dynamically by the `generator_family` column.

Errors, noise, redundancies? Documented in Section 4. Per-agent fail rates and κ versus the human gold sheet are released alongside the corpus. The 1,601 B2-flagged questions are kept with disclosure because LLM-judge calibration with humans on closed-book solvability is unreliable ($\kappa \approx 0.007$).

Self-contained, or relies on external resources? Self-contained: question text, supporting facts, and source URLs are all included. Source URLs may rot over time; we mitigate by archiving Tier-1 source pages and recording the access timestamp.

A.3 Collection process

How was the data acquired? By 35 provenance-verified web scrapers (released in `src/scrapers/`). Sources span government registries (INAO, TTB, OIV), inter-governmental bodies, university research groups (notably UC Davis, USDA Extension), Wikipedia, Wikidata, peer-reviewed journals (*OENO One*, *Vitis*, *AJEV*), and curated open datasets. Scraping was rate-limited and used the disclosed user agent `OenoBench-Research/1.0` (academic wine benchmark).

Time frame. Fact extraction: 2026-03 through 2026-04. Question generation: 2026-04 through 2026-05-03. Evaluation: 2026-05-03.

Ethical review. The work is non-IRB-eligible at our institution because (a) no human subjects were enrolled in research, (b) the gold-sheet reviewer is also the lead author, and (c) all source data is public-record information about commercial wine entities. Human review of LLM-generated questions is treated as data-quality assurance, not as human-subjects research.

A.4 Preprocessing / cleaning / labelling

Atomic-fact extraction with entity tagging; per-source paraphrase to avoid verbatim copy; near-duplicate suppression at cosine ≥ 0.92 ; nine-agent automated audit; human gold-sheet rating on a stratified 50-question subset per release.

A.5 Uses

Recommended uses. Evaluation of LLM factual knowledge in the wine domain; ablation studies of bias-mitigation techniques in benchmark construction; study of self-preference effects in LLM-as-judge pipelines; calibration of automated multi-agent QA frameworks against expert human review.

Discouraged uses. Direct deployment as a wine-recommendation or purchasing system without separate alignment, age-gating, and jurisdictional advertising review; fine-tuning on the benchmark itself for ranking purposes (would invalidate the benchmark for the trained model). The scrape-to-audit pipeline should not be retargeted to high-stakes domains (medicine, law) without comparable expert human review of the retargeted source taxonomy.

Alcohol context. Wine is an alcoholic beverage. OenoBench is a knowledge-evaluation tool, not a consumer-facing product. A high OenoBench score reflects factual / reasoning capability about wine; it does not endorse downstream deployment as a drinking-recommendation system, which would require separate alignment for health information, age-gating, and jurisdictional advertising rules.

A.6 Distribution

Will the dataset be distributed? Yes, under **CC-BY-SA-4.0** on HuggingFace at <https://huggingface.co/datasets/oenobench/oenobench>. A Croissant manifest is included [1]. The construction code is on GitHub at <https://github.com/nikitahudov/oenobench>.

Citation.

```
@inproceedings{khudov2026oenobench,
  title = {OenoBench: A Wine-Domain Benchmark for Knowledge-Grounded
    Evaluation of Large Language Models},
  author = {Khudov, Nikita},
  booktitle = {Advances in Neural Information Processing Systems
    (NeurIPS), Datasets and Benchmarks Track},
  year = {2026}
}
```

A.7 Maintenance

Maintainer. Nikita Khudov, nikitahudov@gmail.com.

Update cadence. Wine regulation changes on a multi-year cadence; we plan annual re-extraction. Scraper code is released so the community can re-extract independently. Corpus versioning follows `release_vX.Y` (major release / minor patch) with all prior versions retained on HuggingFace.

Errata. A public errata log will be maintained alongside the HuggingFace release; corrections to questions, facts, audit verdicts, or difficulty labels will be versioned and dated. Reports of factual errors are accepted via GitHub issues and will be evaluated against the source URL.

B Additional Methodology Detail

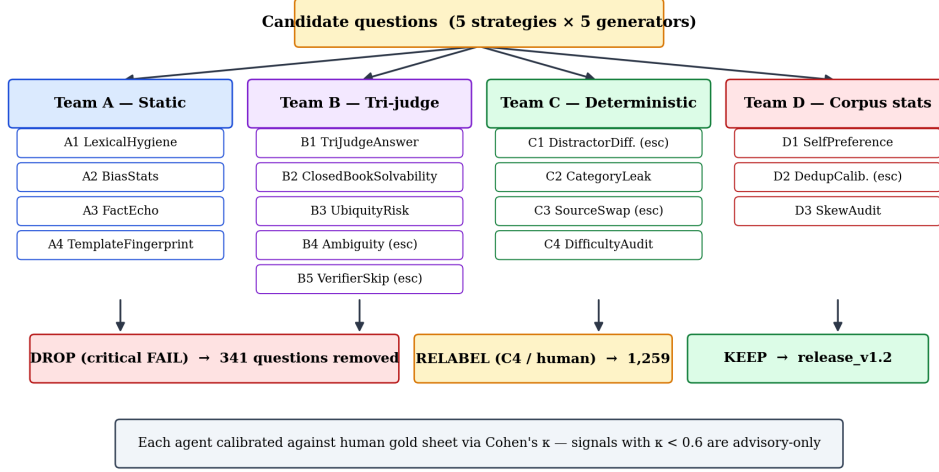


Figure 5: Multi-agent audit architecture. Candidate questions are fanned out to four teams; each team independently emits per-question verdicts. Critical FAIL signals route the question to the drop pool; C4 difficulty disagreements route to relabel; otherwise the question is kept in `release_v1.2`. Every agent is calibrated against the human gold sheet via Cohen’s κ .

B.1 Atomic fact extraction pipeline

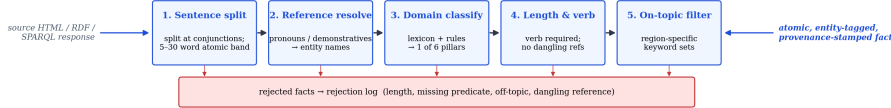


Figure 6: Atomic-fact extraction pipeline. Inputs are heterogeneous (HTML pages, RDF graphs, SPARQL responses, CSV dumps); outputs are uniform single-assertion sentences with entity tags, domain label, source URL and tier.

The atomic-fact extractor (`src/scrapers/_fact_processing.py`) applies five sequential filters to each candidate sentence:

1. **Sentence split.** NLTK Punkt tokeniser; further split at explicit conjunctions (“and”, “but”, “while”) when both clauses have an independent verb. Maximum atomic length 30 words; reject longer.
2. **Reference resolution.** Pronouns and demonstratives are replaced with their entity referent from article context (paragraph-level coreference using a rule-based resolver tuned on Wikipedia leads). Sentences with unresolved *it/this/that* are dropped.
3. **Domain classification.** Lexicon-based + rule-based classifier maps each fact to one of the six domain pillars; ambiguous facts (no anchor term in any domain lexicon) are dropped.
4. **Length and predicate validation.** Reject facts <5 words, >50 words, missing a verb, or carrying dangling references / fragments. The 5–30 word band is the strict filter; the 31–50 band is kept with reduced confidence (910 facts).

5. **On-topic filter.** Region-specific keyword sets prevent cross-contamination (e.g. Austrian content in a Bordeaux scraper). Fail $\Rightarrow$ drop.

A 2026-04 audit found that 19 of an original 35 scrapers contained hard-coded LLM-generated facts disguised as scraped data. The fix required rebuilding all 19 against live sources; the lesson preserved for future retargeting is that *provenance auditing must run on the scrapers themselves*, not only on the collected facts.

B.2 Wikidata SPARQL: P17 vs P131*

A common bug in early scrapers used the transitive administrative-parent property (P131*) to bind questions to a country. This introduced severe cross-region contamination, e.g. Austrian appellations appearing in a Bordeaux scraper because the SPARQL graph traversal walked through shared ancestor entities. Switching to the direct country relation (P17) eliminated cross-country leakage at the cost of $\sim 3\%$ lost coverage on borderline-region entities. The released SPARQL templates use P17 exclusively and we recommend the same discipline for any pipeline retargeted to a domain with hierarchical political subdivisions.

B.3 Question-generation prompts

The five strategies share a common JSON output schema enforced via Pydantic with a 3-tier extraction fallback (markdown-fenced, raw JSON, prefix-pattern). Per-strategy prompt summaries:

- **Fact-to-question:** "Given a single atomic wine fact, write a 4-option multiple-choice question whose unique correct answer is grounded in the fact. Paraphrase the fact; do not copy contiguous spans of >5 words. Distractors must be confusable same-category entities, not random alternatives."
- **Comparative:** given two related facts (matched on entity type, country/sub-domain), "ask which differs in the named attribute or what both share. Avoid iconic-vs-non-iconic pairings."
- **Scenario synthesis:** given a cluster of 3–5 facts, "write a brief domain-appropriate professional scenario (sommelier service / winemaker decision / viticulturist call / business/regulatory choice) with a unique correct answer that requires *all* the cluster facts to derive."
- **Distractor mining:** given a fact + 3 confusable entities, "write a question whose 3 wrong options are these entities, ranked by ascending plausibility."
- **Template:** 45 deterministic parameter substitution templates; no LLM call. A subset is paraphrased post-hoc by Gemini Pro to defeat the A4 stylometric fingerprint detector.

Full prompt texts are in `src/generators/_prompts.py`.

B.4 Closed-book gate specification

The closed-book pre-screen (`src/generators/_closed_book_gate.py`) runs a per-difficulty model:

- L1: Claude Haiku 4.5 reads the question with no source fact. Correct answer $\Rightarrow$ tag `closed_book_solvable`.
- L2: Claude Sonnet 4.6 same protocol.
- L3: Claude Opus 4.7 same protocol; the corpus quota cap fires here.
- L4: gate is skipped by protocol; expert items are expected to be answerable only with the source fact.

The gate's role in generation is to enforce a corpus-level cap on closed-book-solvable items; early pilots used a 25% cap, but it was **raised to 50%** after the gold-sheet calibration revealed that the LLM panel over-attributes closed-book solvability relative to humans (Section 4.2), so the higher cap retains the items that are in fact discriminating in evaluation. The audit's B2 agent later re-runs the check at corpus scale on a tri-judge panel. Failures are not dropped; they are reserved into the closed-book slice and used for the analytic partition in Section 5.5.

B.5 Audit agent specifications

Each agent is a versioned Python module under `src/qa/agents/`. Agents emit (`question_id`, `agent_id`, `version`, `severity`, `payload_json`) tuples to the `audit_findings` table; `severity` is one of {PASS, WARN, FAIL}. Versioning is strict: the (`run_id`, `question_id`, `agent_id`, `version`) tuple is unique, so re-running an agent at a new version produces additive findings rather than overwriting. We document each agent’s threshold and gold-sheet κ here.

Table 4: Audit-agent thresholds and gold-sheet calibration.

Agent	Signal	Threshold	κ vs. gold
A1 LexicalHygiene	vague/marketing regex hits	≥ 1 hit $\Rightarrow$ WARN; $\geq 3 \Rightarrow$ FAIL	0.71
A2 BiasStats	χ^2 position; M–W length	$p < 0.01 \Rightarrow$ FAIL	0.65
A3 FactEcho	LCS / fact length	$\geq 0.45 \Rightarrow$ WARN; $\geq 0.65 \Rightarrow$ FAIL	0.83
A4 TemplateFingerprint	POS-bigram logreg held-out AUC	> 0.84 corpus-level $\Rightarrow$ WARN	0.62
B1 TriJudgeAnswer	majority disagrees with key	2/3 disagree $\Rightarrow$ FAIL	0.78
B2 ClosedBookSolv.	majority solves no-source	2/3 solve $\Rightarrow$ WARN; 3/3 $\Rightarrow$ FAIL	0.007
C2 CategoryLeak	wine-type mismatch in distractors	any $\Rightarrow$ FAIL	0.91
C4 DifficultyAudit	$ \Delta $ assigned vs. rated	≥ 1 WARN; ≥ 2 FAIL	0.74
B3 UbiquityRisk	ubiquitous-grape stem $\times$ region answer	both $\Rightarrow$ WARN	0.69

The B2 row is the calibration finding discussed in Section 4.2: the LLM panel and the human reviewer disagree on what counts as “world-knowledge solvable”, because frontier LLMs have absorbed substantial wine knowledge during pre-training. We retain the agent as a measurement of the LLM panel’s prior, and report the contextual / closed-book partition (Section 5.5) that turns the disagreement into a useful diagnostic.

Release-cycle audit findings. Table 5 summarises the verdicts produced when the 9-agent audit was applied to `release_v1.1` (3,670 candidate questions); these are the numbers cited in Section 4.3.

Table 5: Audit findings on `release_v1.1` and the resulting drop / re-label policy.

Audit signal	Triggered FAIL	Action
A1 LexicalHygiene (vague phrasing)	60	drop
A3 FactEcho (verbatim copy)	63	drop
B1 TriJudgeAnswer (wrong answer key)	47	drop
C2 CategoryLeak (wine-category leak)	9	drop
B3 UbiquityRisk (grape $\times$ region ambiguity)	183	drop
Distinct dropped (with overlap)	341	
C4 DifficultyAudit (relabel from 1,252 LLM hits + 7 human)	1,259	re-label
B2 ClosedBookSolvability (kept w/ disclosure)	1,601	disclose

Difficulty distribution before and after C4 re-label. Table 6 shows the corpus-level effect of the C4 difficulty re-label, which shifted the corpus from 14% L3+L4 items to 51% L3+L4.

Table 6: Difficulty distribution before and after C4 re-label.

	Generator-assigned	Post-relabel	Δ
L1 (entry)	1,261 (38%)	694 (21%)	−567
L2 (intermediate)	1,559 (47%)	894 (27%)	−665
L3 (advanced)	218 (7%)	678 (21%)	+460
L4 (expert)	291 (9%)	1,001 (31%)	+710

B.6 Calibration philosophy

We emphasise the design choice underlying these numbers. A naive “automated audit” interpretation would have dropped all 1,601 B2-flagged questions, which would have removed nearly half the corpus and biased the remainder toward whatever idiosyncratic items the LLM panel happened not

to know. By calibrating B2 against humans and treating it as an evaluator-side signal, we preserve dataset value while giving downstream users a principled way to interpret leakage. The audit’s role, on this reading, is not to be the final arbiter of quality, but to surface signals *that humans validate* as quality-relevant, and to disclose clearly the signals where humans and LLMs disagree.

C Full Evaluation Tables

We give the full per-domain, per-strategy, and per-difficulty breakdowns referenced in Section 5, plus the cost ledger and per-config self-preference detail.

C.1 Per-domain breakdown

Table 7: Per-config per-domain accuracy (%).

Config	Regions	Varieties	Producers	Viticulture	Winemaking	Business
claude-opus-4.7	83.2	80.2	86.8	77.9	86.6	64.2
claude-haiku-4.5	55.8	49.7	58.7	54.2	56.7	37.8
gpt-5	88.1	78.2	88.4	82.8	80.7	63.0
gpt-5-mini	82.1	74.4	83.5	77.5	77.5	66.3
gemini-2.5-pro	87.0	78.8	89.0	78.5	77.5	61.8
gemini-2.5-flash	76.9	72.9	82.3	73.6	77.5	60.6
llama-3.3-70b	68.9	62.9	77.2	65.7	70.6	50.4
llama-3.1-8b	62.2	55.6	69.5	61.7	59.4	47.6
deepseek-v3	74.4	68.5	73.2	70.8	69.0	51.2
qwen-2.5-72b	69.0	65.1	73.4	67.5	70.1	52.8
qwen-2.5-7b	59.1	55.6	59.4	57.4	57.8	44.7
mistral-large-2411	71.5	65.6	75.4	69.4	73.3	51.6
o3	88.4	81.3	90.9	80.3	76.5	65.4
gemini-2.5-pro-thinking	86.5	80.5	90.0	80.5	83.4	60.2
deepseek-r1	80.5	73.6	85.6	72.2	76.5	64.6
claude-opus-4.7-thinking	83.2	80.9	87.0	78.7	84.0	61.4
Mean (16 configs)	76.0	70.2	79.4	71.8	73.6	56.5

C.2 Per-strategy breakdown

Table 8: Per-config per-strategy accuracy (%).

Config	FTQ	Scenario	Template	Comparative	Distractor
claude-opus-4.7	77.7	85.6	92.8	79.5	82.7
claude-haiku-4.5	47.5	62.7	70.7	53.3	56.8
gpt-5	79.9	80.9	95.4	79.5	87.9
gpt-5-mini	73.8	80.6	92.8	78.3	84.7
gemini-2.5-pro	77.4	87.5	93.1	82.4	86.4
gemini-2.5-flash	69.5	84.0	89.7	77.5	79.5
llama-3.3-70b	59.5	80.6	89.7	64.8	71.9
llama-3.1-8b	52.2	70.5	89.7	60.2	63.7
deepseek-v3	64.5	80.3	86.4	71.7	73.3
qwen-2.5-72b	59.6	79.3	88.7	69.3	73.6
qwen-2.5-7b	49.1	72.1	78.9	57.0	61.0
mistral-large-2411	62.7	80.6	87.4	67.6	73.3
o3	79.9	84.3	94.9	84.8	88.4
gemini-2.5-pro-thinking	78.5	88.4	95.6	84.0	84.2
deepseek-r1	73.4	75.5	92.5	77.0	80.7
claude-opus-4.7-thinking	77.3	85.6	93.6	80.3	83.0
Mean (16 configs)	67.6	79.9	89.5	73.0	76.9

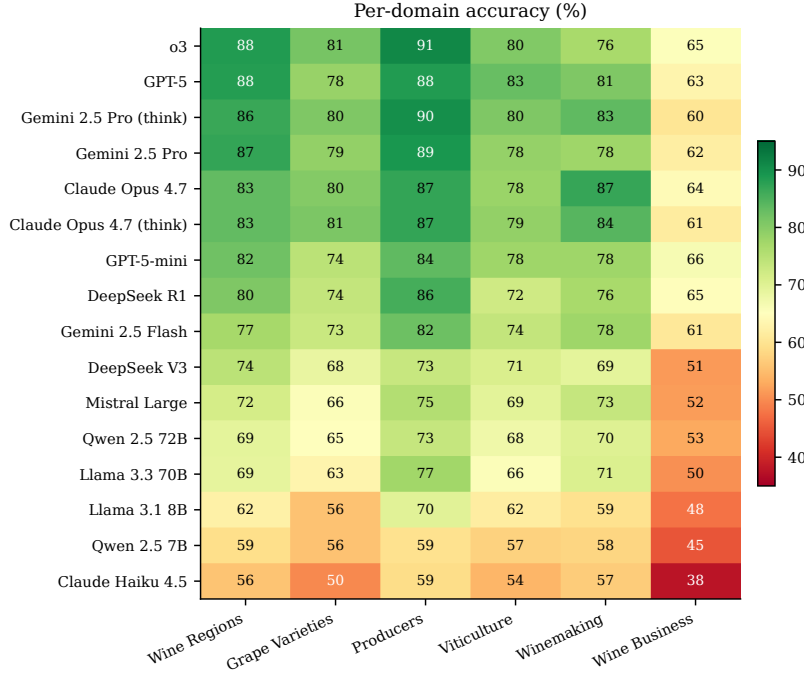


Figure 7: Per-domain accuracy as a heatmap. *Wine business* is the hardest domain across every tier (mean 56.5%) — both because business-domain facts are sparser in the training distribution and because the C4 difficulty re-label promoted business questions that are computationally harder (multi-fact reasoning over regulation).

Table 9: Per-config per-difficulty accuracy (%).

Config	L1 (entry)	L2	L3	L4 (expert)
claude-opus-4.7	98.0	77.0	86.7	69.1
claude-haiku-4.5	74.3	48.3	59.9	38.8
gpt-5	98.3	82.4	87.5	69.2
gpt-5-mini	98.0	73.7	84.7	64.8
gemini-2.5-pro	98.6	79.2	87.3	68.5
gemini-2.5-flash	95.4	70.9	82.2	60.1
llama-3.3-70b	91.9	60.6	74.8	50.3
llama-3.1-8b	84.8	55.4	65.3	45.0
deepseek-v3	95.8	65.9	76.3	52.4
qwen-2.5-72b	93.2	60.5	73.7	51.4
qwen-2.5-7b	82.0	50.6	60.0	43.3
mistral-large-2411	94.4	63.1	73.3	54.0
o3	98.4	81.4	89.4	71.2
gemini-2.5-pro-thinking	99.1	80.2	88.3	69.4
deepseek-r1	97.5	74.2	82.0	62.1
claude-opus-4.7-thinking	98.3	76.8	87.0	68.6
Mean (16 configs)	93.6	68.8	78.7	58.7

C.3 Per-difficulty breakdown and reasoning lift

On the L2/L3/L4 means in Table 9. Two features of the column averages deserve comment. First, the L2 mean (68.8%) is *below* the L3 mean (78.7%) — an apparent inversion of the difficulty ordering. This is an artefact of the C4 difficulty re-label (Section 4): many questions originally labelled L2 by their generator were promoted to L3, leaving behind a residual L2 set that is concentrated in the *wine_business* pillar (the hardest domain by a wide margin, Table 7). The L3 set, in contrast, was enriched with cleanly-stated multi-fact items from the higher-quality scrapers. Second, the L3 → L4 drop is real and large (78.7% → 58.7%, a 20 pp fall), reflecting that L4 is the expert tier where

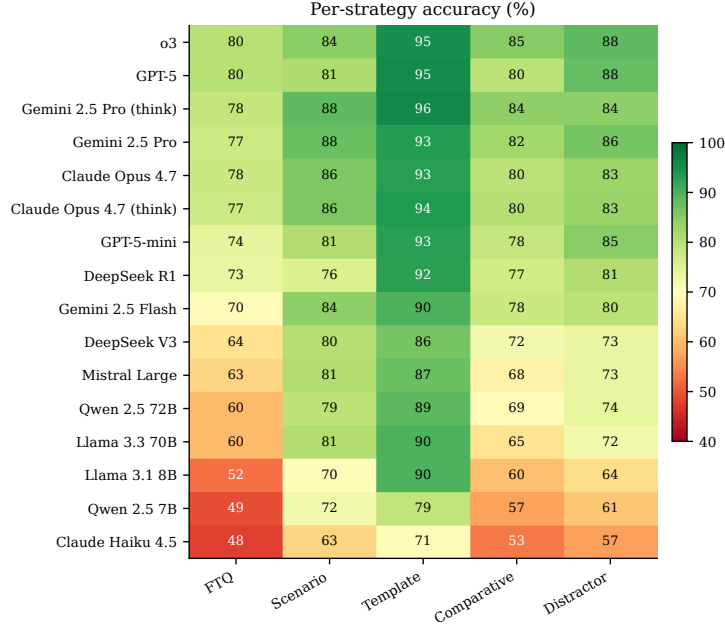


Figure 8: Per-strategy accuracy heatmap. Templates are the easiest strategy uniformly (89.5% mean), even for small models, because they are deterministic single-fact recall questions. The fact-to-question strategy is hardest because its distractors are sampled from confusable same-category entities; comparative questions add the constraint that the model must integrate two facts.

the source fact is genuinely required to answer the question; that fall is also where the inter-config spread is widest (Section 5.1), making L4 the most discriminating slice for distinguishing models. Both effects are intended outcomes of the audit-driven difficulty re-label rather than miscalibration.

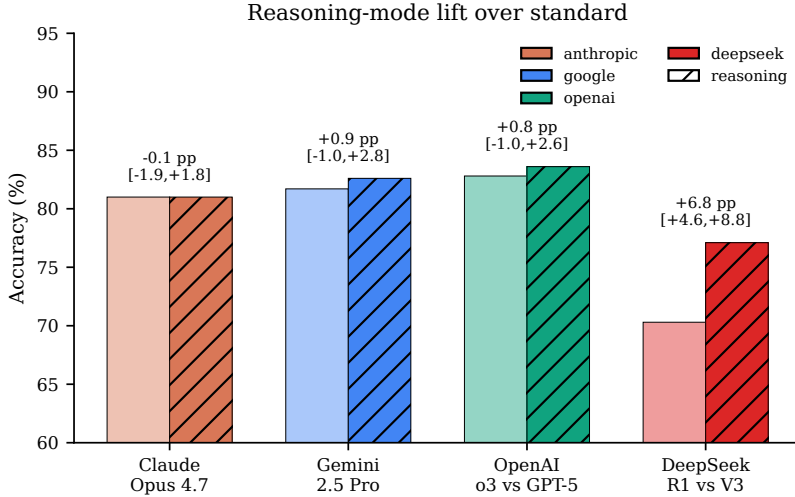


Figure 9: Reasoning-mode vs. standard accuracy for four families; δ in pp with 95% bootstrap CI (1,000 resamples). Only DeepSeek’s CI excludes zero.

C.4 Self-preference bootstrap CIs

C.5 Closed-book vs. source-grounded performance

Table 10: Reasoning lift per difficulty tier (δ = thinking – standard, pp).

Pair	L1	L2	L3	L4
Claude Opus 4.7 (think vs std)	+0.3	−0.2	+0.3	−0.5
Gemini 2.5 Pro (think vs std)	+0.5	+1.0	+1.0	+0.9
OpenAI o3 vs GPT-5	+0.1	−1.0	+1.9	+2.0
DeepSeek R1 vs V3	+1.7	+8.3	+5.7	+9.7

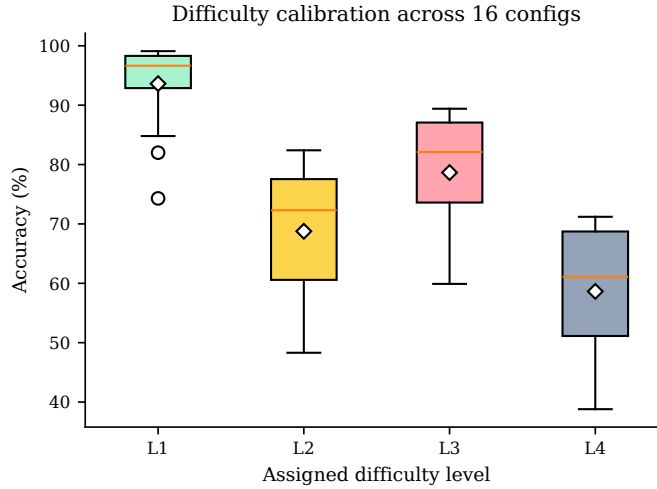


Figure 10: Difficulty calibration: distribution of accuracy across the 16 configs at each level. The medians sit at L1 96%, L2 70%, L3 81%, L4 60%.

Table 11: Per-config Self-Preference Score with 95% bootstrap CI (1,000 resamples). DeepSeek and Mistral have no own-family questions in release_v1.2 and are omitted.

Config	Family	own (%)	other (%)	δ (pp)	95% CI
claude-haiku-4.5	anthropic	58.8	48.8	+10.0	[+5.7, +14.5]
claude-opus-4.7	anthropic	86.6	77.5	+9.1	[+6.1, +12.1]
claude-opus-4.7-thinking	anthropic	86.9	77.2	+9.7	[+6.5, +13.1]
qwen-2.5-7b	qwen	60.9	51.9	+9.0	[+4.5, +13.2]
qwen-2.5-72b	qwen	66.1	64.1	+2.0	[−2.1, +6.0]
llama-3.3-70b	meta	65.5	63.6	+1.9	[−2.3, +6.1]
gpt-5-mini	openai	77.9	76.1	+1.7	[−2.4, +5.4]
o3	openai	81.9	82.1	−0.1	[−3.9, +3.3]
gpt-5	openai	79.7	81.4	−1.7	[−5.4, +2.1]
llama-3.1-8b	meta	54.2	57.2	−3.0	[−7.3, +1.4]
gemini-2.5-pro	google	75.0	81.1	−6.1	[−10.7, −1.8]
gemini-2.5-pro-thinking	google	74.0	82.0	−8.0	[−12.3, −3.8]
gemini-2.5-flash	google	64.3	74.7	−10.4	[−15.3, −5.6]

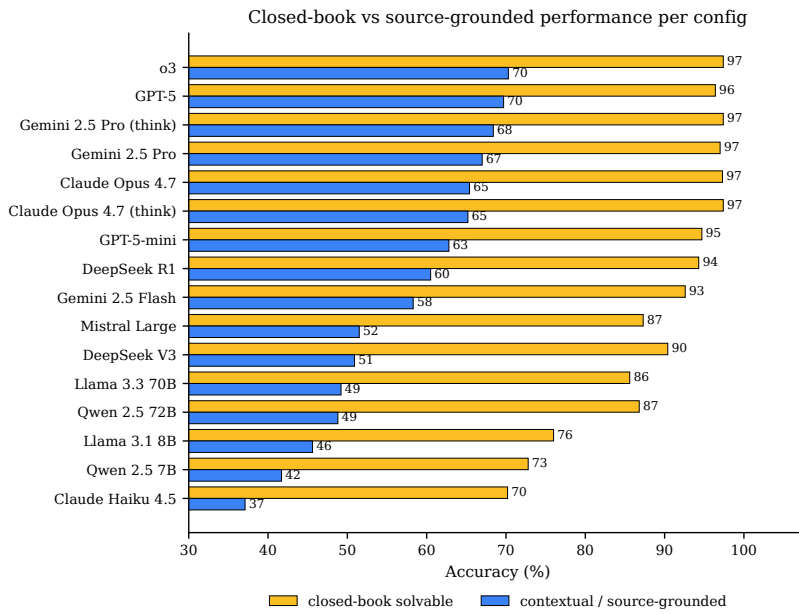


Figure 11: Per-config accuracy on B2-flagged closed-book-solvable questions ($n=1,601$) vs. contextual / source-grounded questions ($n=1,665$); every config gains +26.6 to +39.6 pp on the closed-book slice.

D Application Screenshots

D.1 Human-review web application

The human-review web application is the front-end through which WSET-Diploma reviewers rate stratified gold-sheet questions and the through which the agent-vs-human κ values reported in Section 4.2 are collected. It is a Flask application with two layers of authentication (outer HTTP Basic for shared-link access, inner per-reviewer session cookies) and writes ratings into a separate `human_reviews` Postgres table to keep gold-sheet labels independent of automated audit findings. Each reviewer is associated with a WSET certification level recorded at registration, and the app supports multi-reviewer inter-rater agreement (κ) computation out of the box.

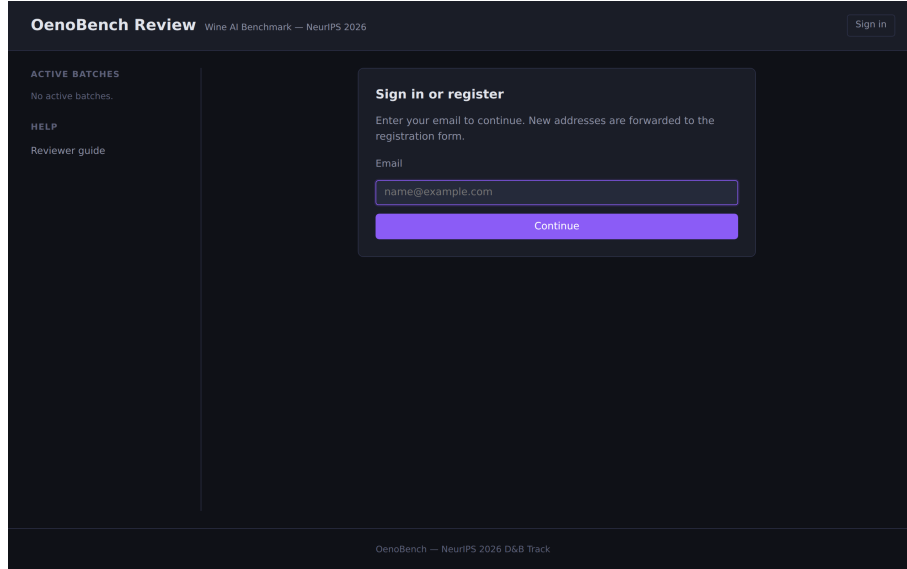
The screenshot shows a web application interface with a dark theme. At the top, a header bar contains the text "OenoBench Review" followed by "Wine AI Benchmark — NeurIPS 2026" and a "Sign In" button on the right. On the left side, there is a sidebar menu with the following items: "ACTIVE BATCHES" (with a sub-item "No active batches."), "HELP", and "Reviewer guide". The main content area is dominated by a "Sign in or register" form. This form includes the instruction "Enter your email to continue. New addresses are forwarded to the registration form." Below this is an "Email" label and a text input field containing "name@example.com". At the bottom of the form is a blue "Continue" button. The footer of the page, located at the bottom center, reads "OenoBench — NeurIPS 2026 D&B Track".

Figure 12: Sign-in / register screen. Outer HTTP Basic Auth gates shared-link access; the inner form establishes a per-reviewer session for IRR attribution.

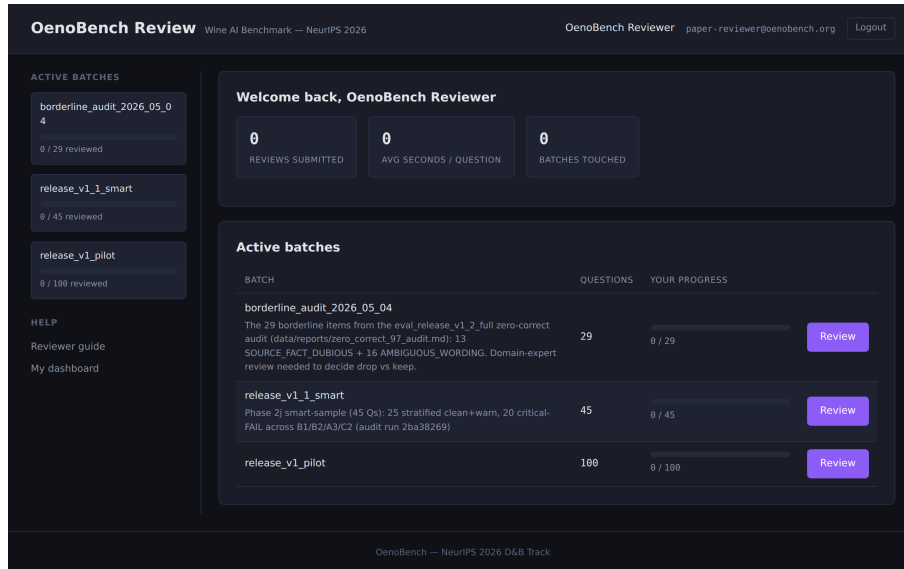


Figure 13: Reviewer dashboard. Active batches are listed; clicking a batch enters the rubric-scoring flow (an eight-rubric grid with PASS / WARN / FAIL per rubric, inline guidance from the reviewer guide, and per-submission time-on-page tracking). The reviewer guide is linked permanently in the sidebar.

E Released Artifacts and Reproducibility

E.1 Released artifacts

- **Dataset (HuggingFace).** <https://huggingface.co/datasets/oenobench/oenobench> — release_v1.2, 3,266 questions, Parquet test split, README.md datasheet, croissant.json manifest. License: CC-BY-SA-4.0. Versioning is permanent: prior releases remain accessible via revision tags.
- **Code (GitHub).** <https://github.com/nikitahudov/oenobench> — full pipeline: 35 scrapers, 5 question generators, 14 audit agents, the human review web app, and the monitoring dashboard. Apache 2.0.
- **Process log.** docs/PROCESS_LOG.md — chronological lab-notebook of every phase shipped, with sources, methodology, quality-control counts, and decision rationale, in the format prescribed for the methodology sections of this paper.
- **Audit reports.** docs/QUALITY_AUDIT_REPORT.md, docs/RELEASE_V1_2_AUDIT_ACTIONS.md, docs/GOLD_CALIBRATION_ANALYSIS.md, data/reports/zero_correct_97_audit.md.
- **Eval reports.** data/reports/eval_release_v1_2_full_cleaned.md (the 16-config × 3,266-Q run reported in this paper).

E.2 Reproducibility scripts

The construction pipeline is driven by a small number of shell scripts in scripts/:

- `run_all_scrapers.sh` — orchestrates the 35 scrapers end-to-end. Each scraper writes a timestamped log to data/logs/.
- `run_release_v1_build.sh` — full generation pipeline invocation; idempotent and resume-safe via stamped tags.
- `export_release_v1_2_to_parquet.py` — produces the HuggingFace Parquet directly from Postgres; the released file is the byte-identical output of this script.
- `build_smart_review_sheet.py` — generates a 50-question stratified gold-sheet review batch (random + critical-FAIL + borderline-WARN strata).

- `tag_audit_actions.py` — applies the audit drop / relabel policy to a tagged corpus; this is the script that produced the `release_v1.2` state from `release_v1.1`.

E.3 Database schema

The released schema is initialised by the migrations under `config/postgres/` (six idempotent files, applied in order on container startup): `init.sql` (sources, facts, questions, question_facts, generation_metadata, evaluation_runs, evaluation_answers, human_reviews), `002_audit_schema.sql` (audit_runs, audit_findings, audit_gold_labels, audit_severity enum), `003_sample_schema.sql` (mirror schema for the Phase 5 sample-DB eval), `003_cb_reserve.sql` (closed-book reserve pool tagging), `004_eval_telemetry.sql` (provider, tokens, reasoning_config, latency columns on evaluation_answers), `005_or_cost_telemetry.sql` (OpenRouter authoritative cost columns).

The released schema is documented in code (`config/postgres/`); the load-bearing tables are sources (registry), facts (atomic facts grounded by source), questions + question_facts (corpus and fact-question links), generation_metadata (per-question generator and strategy), audit_findings (per-agent verdicts), and human_reviews (gold-sheet ratings).

E.4 Compute and cost

The full evaluation reported in Section 5 ran in **120 minutes 34 seconds** on a single VM driving 16 OpenRouter configurations in parallel. **Total LLM API spend across the entire project was \$783**, paid out-of-pocket to OpenRouter and covering all generation pilots, audit cycles, gold-sheet calibration runs, and the final 16-config evaluation. No GPU compute was used; all model inference was via OpenRouter (a unified API gateway). The release artefacts include all per-call telemetry (provider, tokens, reasoning configuration, p50/p95 latency, OR-authoritative cost) for the full evaluation run; the per-config cost ledger is reproduced inline as Table 12.

Table 12: Per-config cost ledger for the full 16-config evaluation (3,266 questions per config). Effective OpenRouter spend.

Config	Total cost	Correct ans.	Cost / correct
llama-3.1-8b	\$0.01	1,976	\$0.0000
llama-3.3-70b	\$0.04	2,190	\$0.0000
gemini-2.5-flash	\$0.12	2,454	\$0.0000
deepseek-v3	\$0.12	2,295	\$0.0001
qwen-2.5-7b	\$0.12	1,860	\$0.0001
qwen-2.5-72b	\$0.15	2,202	\$0.0001
claude-haiku-4.5	\$0.44	1,741	\$0.0003
mistral-large-2411	\$0.87	2,256	\$0.0004
gpt-5-mini	\$2.82	2,561	\$0.0011
claude-opus-4.7	\$3.35	2,647	\$0.0013
claude-opus-4.7-thinking	\$3.53	2,645	\$0.0013
deepseek-r1	\$10.54	2,517	\$0.0042
o3	\$11.80	2,729	\$0.0043
gemini-2.5-pro-thinking	\$13.06	2,698	\$0.0048
gpt-5	\$21.90	2,704	\$0.0081
gemini-2.5-pro	\$29.47	2,669	\$0.0110

E.5 Test status

771 of 771 unit tests pass on main as of the camera-ready build. Tests cover the fact-processing pipeline, the five generation strategies, all 14 audit agents, the orchestrator, the closed-book gate, the eval harness (slot dispatch, cost computation, reasoning-mode prompt assembly), and the review-app endpoints.

F Construction Timeline

The full chronological lab notebook of construction is in `docs/PROCESS_LOG.md`. We summarise the major dated milestones here for paper-traceable provenance.

Table 13: Construction-and-evaluation timeline (selected milestones).

Phase	Dates	Milestone
1	2026-03-12 – 04-01	35 scrapers built; first 16,702-fact corpus assembled.
1.1	2026-04-07	<i>Provenance audit reveals 19 scrapers contained hardcoded LLM-generated facts</i> ; corpus paused.
1.2	2026-04-08 – 04-11	Shared infrastructure (<code>_fact_processing.py</code> , <code>_web_helpers.py</code> , <code>_wiki_helpers.py</code>) shipped; Phase 1 rebuild fixes 8 quality-issue scrapers; Phase 2 rebuild replaces all 17 hardcoded scrapers with genuine Wikipedia + SPARQL + official sources. <i>38,104 fact corpus restored, all provenance-traceable</i> .
2	2026-04-12 – 04-22	Five generation strategies built and iteratively tuned; multi-model orchestrator with quota management; closed-book gate v2 wired into all five generators.
2c	2026-04-19 – 05-02	Nine-agent audit framework shipped across 4 teams; sixteen audit pilots (v1–v16) tune thresholds and speedups; 472/472 → 771/771 tests pass.
2g	2026-04-28 – 05-02	Ten generation-pipeline speedup levers shipped (LLM cache, in-process dispatch, Thread-Pool dispatch, substantiveness filter, circuit breaker, tier-aware gate). v9 build wall: 18 min vs. v8’s 11.4 h.
2j	2026-05-03	<code>release_v1.1</code> (3,670 candidate questions) assembled from staged generation runs and the curated sample-DB. Audit cycle drops 341 questions on critical FAILs and re-labels difficulty for 1,259 (51% L3+L4 vs. prior 14%). Cycle cost \$76 / 5h 22m.
4	2026-05-03	Human-review web app shipped on port 5556 with separate <code>human_reviews</code> schema and multi-reviewer IRR support.
5	2026-05-03 – 05-04	Full evaluation: 16 configs × 3,329 questions in 120m 34s for \$98.33; post-eval audit of 97 zero-correct items removes 54 defects + 9 borderline-review drops; <code>release_v1.2</code> locked at 3,266 questions.
HF	2026-05-04	Dataset card + Croissant manifest + Parquet test split published; this paper drafted.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [\[Yes\]](#)

Justification: Section 1 lists four contributions, and each is supported by a numbered section: dataset construction (Section 3), automated audit (Section 4), bias-aware evaluation (Section 5). The abstract’s headline numbers (3,266 questions, 16-config eval, +32.6 pp closed-book vs. contextual gap) are reproduced with full tables in Appendix C.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [\[Yes\]](#)

Justification: Section 7 discusses six limitations explicitly, including English-only scope, geographic skew, snapshot vs. moving target, B2 audit calibration ceiling, residual self-preference, and single-reviewer gold sheet.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [\[N/A\]](#)

Justification: This is a dataset-and-benchmark paper; no formal theoretical results are claimed.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper?

Answer: [\[Yes\]](#)

Justification: Appendix E lists the construction scripts, the dataset URL, the model slate, the per-call telemetry available alongside the released corpus, and the deterministic build pipeline. Section 5 reports run id, wall time, total cost, and exact config strings.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: Code at <https://github.com/nikitahudov/oenobench> (Apache 2.0); data at <https://huggingface.co/datasets/oenobench/oenobench> (CC-BY-SA-4.0). Exact reproduction commands are documented in Appendix E.

6. Experimental setting/details

Question: Does the paper specify all the training and test details necessary to understand the results?

Answer: [Yes]

Justification: Section 5 specifies the 16-config slate, single-letter output protocol (A–D, max_tokens=5, five-stop fallback). Appendix C gives full per-config × per-domain × per-strategy × per-difficulty tables. Appendix B gives audit-agent thresholds and prompt summaries.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Reasoning lift, self-preference scores, and closed-book vs. contextual deltas are all reported with 95% bootstrap confidence intervals (1,000 resamples), with the bootstrap procedure documented in Section 5.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources needed to reproduce the experiments?

Answer: [Yes]

Justification: Appendix E reports total wall time (120 min 34 s), total LLM calls (52,256), total cost (\$98.33 effective), and the deployment topology (single VM driving 16 OpenRouter configurations in parallel; no GPU compute).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics?

Answer: [Yes]

Justification: All sources used are public-record; no human subjects research was conducted (the gold-sheet reviewer is the lead author); responsible-use guidance is given in the Datasheet (Appendix A, “Uses”) and in Section 7.

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: The Datasheet (Appendix A) covers alcohol-context and discouraged uses; source-licensing and the non-fabrication guarantee are documented in Section 3.2; personal-data scope and the no-private-individuals constraint are in the Datasheet “Composition”. Section 6 discusses positive applications including specialist-model fine-tuning and pipeline retargeting to other expert domains.

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse?

Answer: [Yes]

Justification: The dataset is text-only multiple-choice questions with public-record entity names; no scraped images or personal-data risks. We document discouraged uses in the Datasheet (Appendix A, “Uses”) including the recommendation against retargeting the pipeline to high-stakes domains without expert review.

12. Licenses for existing assets

Question: Are the creators or original owners of assets used in the paper properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [\[Yes\]](#)

Justification: Source-tier table in Section 3.2 credits all major sources with licenses (Wikipedia CC-BY-SA, Wikidata CC0, INAO Licence Ouverte, TTB public domain, etc.). Per-fact source URLs are released alongside the corpus.

13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [\[Yes\]](#)

Justification: A full Datasheet is in Appendix A; a Croissant manifest [1] ships with the HuggingFace release; all 9 audit agents, 5 generators, and 35 scrapers are documented in source.

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation?

Answer: [\[N/A\]](#)

Justification: No crowdsourcing or third-party human-subjects research. Gold-sheet ratings were produced by the lead author (a WSET-Diploma holder); the review web app screenshots in Appendix D document the rubric and instructions.

15. Institutional review board (IRB) approvals or equivalent for research with human subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether IRB approvals were obtained?

Answer: [\[N/A\]](#)

Justification: As noted in Appendix A, the work involves no human subjects research distinct from data-quality assurance by the lead author.

16. Declaration of LLM usage

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research?

Answer: [\[Yes\]](#)

Justification: LLMs are central to the methodology (multi-strategy question generation across five model families, nine-agent automated audit including a tri-judge panel, closed-book solvability pre-screen). Section 3 and Section 4 describe their usage; Section 3.2 and Appendix B document the explicit fact-grounding constraint that prevents LLMs from being used as the source of factual claims.